

Fig. 4: A diagram showing how a company like Nike can use mass collaboration and 3D printing (Source: <http://madameeureka.wordpress.com>)

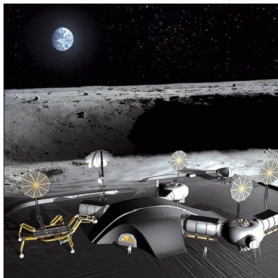


Fig. 5: 3D printed moon base (Source: www.dezeen.com)



Fig. 6: A home 3D Printer and its colourful toy printouts (Source: www.3dsystems.com)

personal 3D printers are also available. And there are both fully assembled and DIY models. One can use the free version of applications such as Google SketchUp and the Blender program with them. Imagine if you and I begin using 3D printing—each of us could ‘manufacture’ something. Today, customers expect a higher level of customisation and personalisation, which

3D printing can facilitate through the concept of mass collaboration. Websites such as Wikipedia, Reddit, etc, are geared up to ‘mass collaborate.’ Reddit is a website which includes various sub-reddits. These sub-reddits enable members to share special interests.

The free software (the starting point in the above diagram) has been developed by Google. A company like Nike is in a position to enlist particular communities that have an interest in design and fashion, as well as access to a 3D printer. Via mass collaboration, people can come together to create new designs and send them to Nike. Personal 3D printers are becoming affordable and free software is also available. This means that home users now have the resources to develop prototypes. This could bring down unemployment and, yes, maybe you can also reinvent your house in a cost effective manner.

Living on the moon: The future of 3D printing

Living on the moon could become a reality with 3D printing. An entire moon base could be printed using lunar dirt as raw material. A team from The European Space Agency (ESA) is working closely with architects to chalk out ideas for a 3D-printed moon colony. The whole idea is to design for extreme climates and exploit the environmental benefits of using local, sustainable materials—just as it is done on Earth.

Using material that exists on the moon would mean there is no need for large and expensive spaceships to

transport materials from Earth. The moon base would be built using a robotic printer moving over an inflatable dome. The base would have hollow walls and a cell-like robust structure that looks like the skeleton of a bird. This would guard humans residing on the moon from gamma radiation and small meteorites. ESA and its partners have simulated building some portion of the base that weighs about a ton using a blend of silicon, aluminium, calcium, iron and magnesium oxides. This is similar to the composition of moon dust. The 3D printer is being run in conditions identical to those on the lunar surface, i.e., the 3D printing is being done inside a vacuum chamber.

The future

Perhaps the evolution of 3D printing could mirror that of photo printing. When printer prices dropped and became affordable, people started printing their own photographs, in the comfort of their homes.

Currently, most people do not print their photos at home. Rather, they actually pay for printing services from high-quality speciality print shops on special occasions. The same scenario may emerge in the case of 3D printing. People will invest in 3D printers as they start to become affordable. Eventually, as 3D print technology becomes omnipresent, users will start approaching specialised shops in order to get quality that stands out from what can be achieved at home. We already have a few companies such as Shapeways, Kraftwurx, Freedom of Creation, etc, that deliver online 3D printing services in both the B2C and B2B space. Customers will upload their 3D designs to the company website, the designs would then be 3D printed using industrial 3D printers and delivered to the customer.

So if you get that 3D printer, you know you can actually do a lot more than just printing your lost hearing aid. ●

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